

MATTHEW D. HALL

EDUCATION

The University of Sheffield

PhD – Automatic Control & Systems Engineering.

OCT 2017 – PRESENT, SHEFFIELD

Lancaster University

MEng 1st class honours – Electrical & Electronic Engineering.

OCT 2012 – JUL 2016, LANCASTER

RESEARCH

Natural Robotics Lab & Sheffield Robotics

Postgraduate Research Student

PROJECT: Self-Reconfiguration of Modular Robots via Active Subtraction.

SUPERVISORS: Dr Roderich Gross (Natural Robotics Lab).

Dr Shuhei Miyashita (Microrobotics Lab).

INTERESTS: Modular Robotics, Reconfigurable Robots, Distributed Control, Industry Applications, Space Exploration.

PUBLICATIONS

M. D. Hall, A. Özdemir, R. Groß, “**Self-Reconfiguration in Two-Dimensions via Active Subtraction with Modular Robots**”, in *Proc. Robotics: Science and Systems XVI (RSS)*, RSS Foundation, 2020.

A. Özdemir, M. Gauci, A. Kolling, M. D. Hall, R. Groß, “**Spatial Coverage Without Computation**”, in *Proc. 2019 International Conference on Robotics and Automation (ICRA)*, IEEE, 2019, pp. 9674–9680.

ORAL PRESENTATIONS

2021: Departmental Postgraduate Research Symposium – *Virtual*.

2020: Int’l. Conf. on Robotics: Science and Systems (RSS) – *Virtual*.

2019: Self-Configuring Modular Robotics for Earth and Space – *York, UK*.

Int’l. Conf. on Robotics and Automation (ICRA) – *Montreal, Canada*.

UK & I Conf. on Transformative Robotics for Industries – *London, UK*.

ACSE Faculty Researcher Symposium – *Sheffield, UK*.

AWARDS & SCHOLARSHIPS

University of Sheffield EPSRC Doctoral Training Partnership National Productivity Scholarship – *The University of Sheffield.*

For course fees and maintenance – 2017–2021

PROGRAMMING SKILLS

- Proficient in Python, C/C++, LaTeX, Git, Shell Scripting.
- Limited experience with MATLAB, Java, HTML, CSS and VHDL.

ACADEMIC EXPERIENCE

Peer Reviewer for International Conferences:

- Autonomous Agents and Multiagent Systems (AAMAS).
- Advanced Networks and Telecommunication Systems (ANTS).
- Distributed Autonomous Robotics Systems (DARS).
- Intelligent Robots and Systems (IROS).
- Multi-Robot Systems (MRS).
- Robotics: Science and Systems (RSS).
- Towards Autonomous Robotic Systems (TAROS).

Graduate Teaching Assistant for Bachelor's and Master's modules:

- Robotics and Autonomous Systems.
- Mobile Robotics and Autonomous Vehicles.
- Robotic Systems.
- Digital and Embedded Systems.
- Foundations of Robotics.

PROFESSIONAL EXPERIENCE

Tourism Advisor – *Use-it Brussels*

JAN 2017 – AUG 2017

- Part time work within a sustainable tourism project, offering tourist information and running tours around Brussels.
- A great opportunity to thoroughly learn about the city and its culture as well as improving my interpersonal skills.

Student Ambassador – *Lancaster University*

OCT 2014 – JUN 2016

- Part-time employment assisting with university events. The role also included the regular running of tours around the university campus.
- These responsibilities required excellent communication and organisational skills, and a personable yet professional demeanour.